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THE WILD FIELDS AND BUDJAK: NOMADS, RIVER FORTS AND BLACK SEA PORTS

Population and Settlement between the Lower Danube and the Seversky Donets

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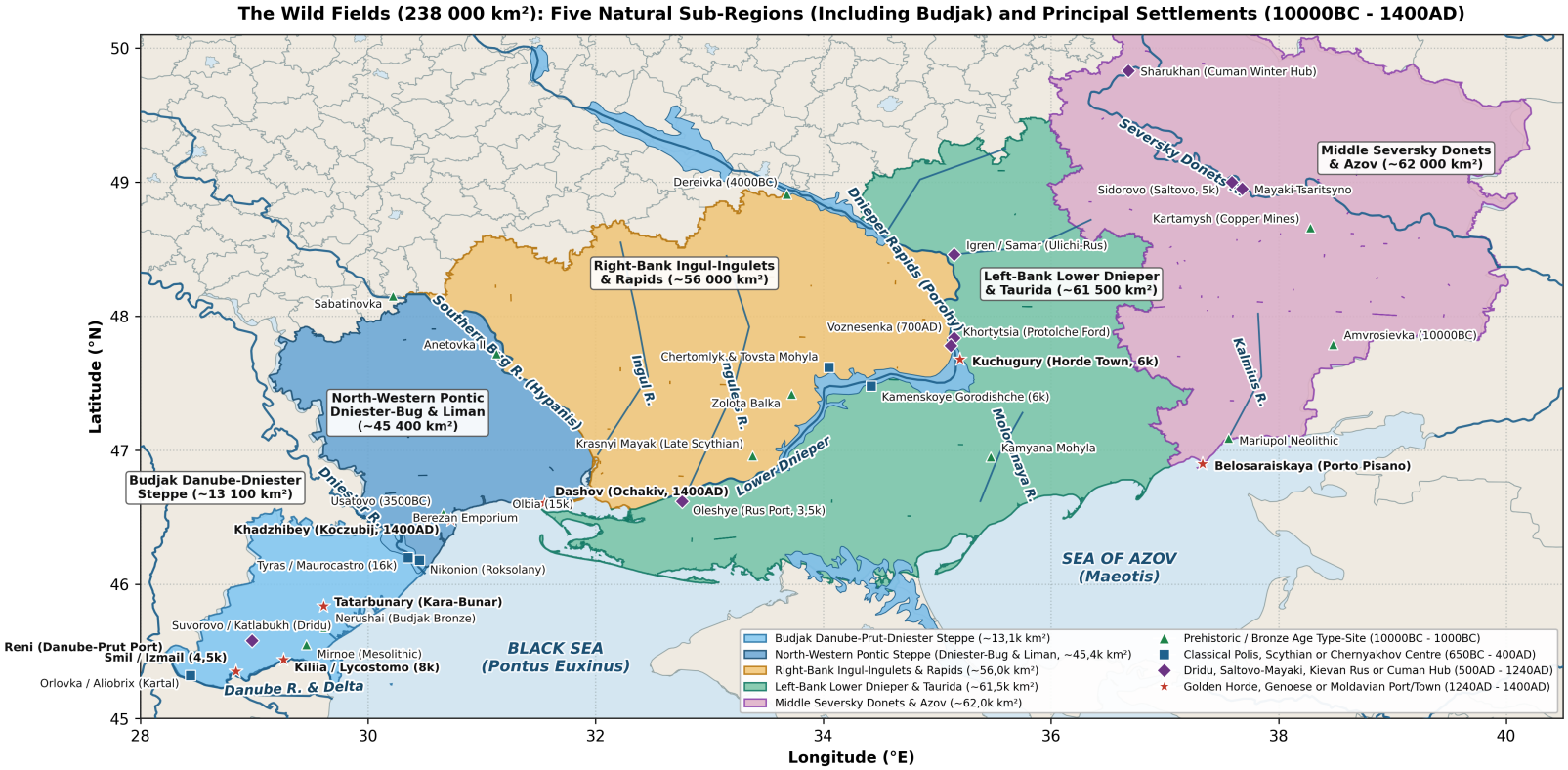


Figure 1. The five regions of the Wild Fields and Budjak (~238k km^2), stretching from the Danube delta and Dniester estuary across the Southern Bug and Lower Dnieper rapids to the Middle Seversky Donets and the Sea of Azov.

Abstract

The Wild Fields (Polish *Dzikie Pola*, Latin *Loca Deserta*, Ukrainian *Dyke Pole* and the western wing of *Desht-i Kipchak*) cover the southern Ukrainian steppe between the Lower Danube and the Seversky Donets, an area of ~238k km^2 north of the Perekop Isthmus and south of the forest-steppe line [1][2]. From small Ice Age and Mesolithic hunting camps along the river valleys (3,8k in 10000BC), farming and herding spread across the steppe in the Copper and Bronze Ages to support 145k people by 1000BC. Greek trading towns on the Black Sea estuaries and Royal Scythian settlements on the Lower Dnieper lifted the population to 235k in 500BC, and the dense farming villages of the Chernyakhov culture reached a Classical peak of 240k in 300AD. After the Huns drove numbers down to 125k by 500AD, Bulgarian *Dridu* villages in Budjak and the Khazar-era Saltovo-Mayaki forts on the Seversky Donets carried the total back to 230k in 900AD, while

trade between Kievan Rus and the Cumans produced a medieval peak of 265k in 1200AD. Although Golden Horde rule revived town life at Aq Qerman, Kiliia and Velyki Kuchugury by 1300AD (210k), the Black Death, Horde civil wars and Timur's 1395AD campaign emptied much of the inland steppe, leaving 155k people in 1400AD, mostly around the Moldavian, Genoese and Lithuanian ports of Budjak and the north-western coast.

Background

Steppe and River Valleys

From 10000BC to 1400AD, settlement in the Wild Fields was shaped by the sharp contrast between the dry upland plains and the river valleys. The open feather-grass steppe (*plakor*) between the major rivers consisted of gently rolling loess plateaus with black-earth and chestnut soils receiving 350 to 480 mm of rain a year; before modern deep wells, these dry watersheds had little surface water in summer and served mainly as seasonal grazing land for nomadic herders [1][3]. By contrast, the deep river valleys, rapids, wooded floodplains (*plavni*), freshwater lakes and coastal estuaries (*limany*) offered shelter and water year-round. Here the Lower Danube (Chilia/Kiliia arm) and its Budjak lakes (Yalpug, Kugurlui, Katlabukh, Kitai and Kagul), the Dniester (*Tyras*), Tiligul, Southern Bug (*Hypanis*), Ingul, Ingulets, Lower Dnieper (*Borysthenes*) with its nine granite rapids (*Porohy*) and the *Veliky Lug* ("Great Meadow") floodplain woods, Samara, Orel, Konka, Molochnaya, Kalmius, Mius and Middle Seversky Donets provided timber, fresh water, rich sturgeon and carp fisheries, winter reed-bed grazing and steep bluffs for defence [4][5][6][7].

Regional Divisions

Following its river basins and natural borders (Figure 1 and Table 1), the 238k km^2 territory divides into five parts.

Region I covers Budjak (~13,1k km^2), bounded by the Lower Danube and Chilia Delta on the south, the Prut confluence and Moldovan border on the west, the Black Sea coast (with the Sasyk, Shagany, Alibey and Burnas lagoons) on the south-east and the Dniester River and Dniester Liman on the north-east. Unlike the inland steppe, Budjak combines an open dry plain (drained by the Kogilnik, Chaga, Sarata and Khadzhider rivers) with two fertile coastal and river zones: the Danube floodplain lakes (Kagul, Yalpug, Katlabukh and Kitai) and the western bank of the Dniester Liman. Because of this position, Budjak long served as the main bridge between the Balkan-Danubian world (Gumelnita farmers, Getae, Roman Lower Moesia, First and Second Bulgarian *Dridu* villages, Byzantium, Genoese *Lycostomo* and Moldavia) and the Pontic steppe (Yamnaya herders, Scythians, Sarmatians, Pechenegs, Cumans and Golden Horde *Aq Qerman*) [8][9][10][11].

Region II spans the north-western Pontic steppe (~45,4k km^2) between the Dniester River and Liman on the south-west, the Kodyma River and forest-steppe line on the north, the Southern Bug (*Hypanis*) and Bug Liman on the north-east and the Black Sea on the south. It contained the Copper Age Usatovo settlements (Usatovo and Mayaki), the Milesian Greek towns and farm villages of Olbia, Berezan, Nikonion and Koshary, Chernyakhov coastal villages (Koblevo and Kamenka-Anchekrak), Tivertsi and Ulichi border settlements and the late fourteenth-century ports of Khadzhibey (*Koczubij*) and Dashov (*Ochakiv*) [12][13][14][15].

Region III covers the right-bank Bug-Dnieper country, the Ingul-Ingulets uplands and the right bank of the Dnieper Rapids (~56k km^2), bounded by the Southern Bug on the west, the forest-steppe border along the Dnieper Upland on the north and the Dnieper (*Borysthenes*) and Dnieper-Bug Estuary on the east and south-east. Taking in the Syniukha, Ingul, Ingulets, Visun, Saksahan and Bazavluk valleys, this region

held the Dereivka Copper Age settlement, the Sabatinovka and Ingul Catacomb Bronze Age heartland, the great Royal Scythian burial mounds (Chertomlyk, Tovsta Mohyla and Alexandropol) and the chain of Late Scythian stone forts along the right bank of the Dnieper (Zolota Balka, Gavrilovka, Krasny Mayak, Konsulovskoye and Nikolaevka-Kazatskoye) [5][16][17].

Region IV comprises the left-bank Lower Dnieper, Veliky Lug, the Samara-Orel basin and the northern Taurida steppe (~61,5k km^2), bounded on the west by the Dnieper, on the north by the Orel (*Oril*) River and forest-steppe border, on the east by the Azov Upland watershed (the Berda divide) and on the south by the Black Sea, Perekop-Sivash and the Sea of Azov. It contained the Vasylivka Mesolithic cemeteries, the Kamyana Mohyla rock sanctuary, the Mikhailovka Yamnaya fort, the 1,2k ha Scythian town of Kamenskoye Gorodishche, the Voznesenka princely camp, the Kievan Rus ports of Oleshye, Protolche (Khortytsia) and Igren-Samar and the Golden Horde towns of Velyki Kuchugury, Mamai-Surka and Konski Vody [18][19][20][21].

Region V takes in the Middle Seversky Donets, the Donets Ridge and the Kalmius-Mius Azov basin (~62k km^2), bounded on the west by the Dnieper-Donets and Azov Upland watershed, on the north-west by the forest-steppe border, on the east by the Aidar-Derkul-Donets-Mius basin divide and on the south by the Sea of Azov. Across the chalk bluffs of the Middle Seversky Donets and Oskol valleys, the Donets Ridge (*Donetskiy Kryazh*) and the Kalmius-Mius coast, this eastern region held the Amvrosievka Palaeolithic hunting camp, the Mariupol Neolithic cemetery, the Bronze Age copper mines of Kartamysh, the eighth- to tenth-century Saltovo-Mayaki forts and towns (Sidorovo, Mayaki-Tsaritsyno, Sukhaya Gomolsha and Mokhnach), the Polovtsian winter towns of Sharukhan, Sugrov and Balin and the Golden Horde and Italian port at Belosaraiskaya (*Porto Pisano*) [6][22][23].

Region	Area (% Share)	Boundaries	Main Settlements & Sites
I. Budjak (Danube-Prut-Dniester Steppe & Liman)	13,1k km^2 (5,5%)	Danube R. & Delta (S), Prut/Moldova (W), Black Sea (SE), Dniester R. & Liman (NE)	Mirnoe, Nerushai, Orlovka (Aliobrix), Tyras/Aq Qerman (Maurocastro), Kiliia (Lycostomo), Smil (Izmail), Reni
II. NW Pontic Steppe (Dniester-Bug & Liman)	45,4k km^2 (19,1%)	Dniester R. (SW), Kodyma & Forest-Steppe (N), Southern Bug R. & Liman (NE)	Usatovo, Mayaki, Nikonion, Berezan, Olbia, Koshary, Sabatinovka, Anetovka II, Khadzhibey, Dashov (Ochakiv)
III. Right-Bank Bug-Dnieper, Ingul-Ingulets & Rapids	56k km^2 (23,5%)	Southern Bug R. (W), Forest-Steppe Line (N), Dnieper R. & Estuary (E/SE)	Dereivka, Chertomlyk, Tovsta Mohyla, Zolota Balka, Gavrilovka, Krasny Mayak, Bashmachka, Voznesensk
IV. Left-Bank Lower Dnieper, Veliky Lug & N. Taurida	61,5k km^2 (25,8%)	Dnieper R. (W), Orel R. & Forest-Steppe (N), Azov Upland Divide (E), Perekop/Azov (S)	Vasylivka, Mikhailovka, Kamyana Mohyla, Kamenskoye, Voznesenka, Khortytsia (Protolche), Oleshye, Kuchugury
V. Middle Seversky Donets, Donets Ridge & Azov Basin	62k km^2 (26,1%)	Dnieper-Donets Divide (W), Forest-Steppe (NW), Aidar-Mius Frontier (E), Azov Sea (S)	Amvrosievka, Mariupol Neolithic, Kartamysh, Sidorovo, Mayaki-Tsaritsyno, Sukhaya Gomolsha, Sharukhan, Sugrov

Total Wild Fields	238k km^2 (100%)	Lower Danube to Seversky Donets & Sea of Azov	191 recorded settlements and sites

Table 1. The five geographical divisions of the Wild Fields and Budjak (238k km^2), showing each region's share of the steppe, its river boundaries and its principal prehistoric, Classical and medieval settlements.

Results

Prehistory and the Bronze Age

Ice Age, Mesolithic and Neolithic River Camps

At the end of the Ice Age (10000BC), as cold steppe gave way to warmer grasslands and rising seas flooded the north-western Black Sea shelf into brackish *liman* estuaries, Upper Palaeolithic and Early Mesolithic hunting bands gathered along limestone river gorges, the Budjak streams and the granite rapids of the Bug and Dnieper [4][14]. Excavations along the Southern Bug (Anetovka II in Region II), the Krynka-Mius basin (the Amvrosievka bison kill site in Region V) and the Lower Dnieper show large group hunts of steppe bison, wild horse and saiga antelope [4]. By the Mesolithic (8000BC), large base camps such as Mirnoe in Budjak (Region I), the sandstone rock sanctuary of Kamyana Mohyla on the Molochnaya River (Region IV) and the cemeteries at the Dnieper Rapids (Vasylivka I-III, Voloske and Igren VIII, with 30 to 50 graves apiece) point to settled fishing communities around the rapids. In 10000BC the regional population stood at about 3,8k (0,3k in Budjak [I], 0,8k in NW Pontic [II], 0,95k in Right-Bank [III], 0,95k in Left-Bank [IV] and 0,8k in Middle Donets [V]), rising to 6,8k by 8000BC.

During the warmer, wetter Neolithic period (6000BC - 5000BC), pottery-making villagers settled along the rivers and estuaries of all five regions. Around Lakes Yalpug, Kagul and Katlabukh in Budjak (Region I, 6000-4500BC), early farmers and fishers in contact with Balkan Starcevo-Cris and Linearbandkeramik groups raised Budjak's population from 1,1k in 6000BC to 1,9k in 5000BC [9][14]. In Regions II and III (6200-5000BC), settlements of the Bug-Dniester Culture clustered along the Southern Bug rapids between Savran, Migeya and Bazkov Ostrov, using flint sickles and keeping early livestock [14]. Across Regions III and IV (6000-4500BC), Sursko-Dnieper and Dnieper-Donets II communities lived on the islands and banks of the Dnieper Rapids (Surskoy Island, Nikolskoye and Vovnigi, with over 130 graves) and around the Samara-Orel confluence [24]. In Region V (5500-4500BC), the 122 red-ochre graves at Mariupol at the mouth of the Kalmius River mark a similar fishing and herding community on the Azov coast. Total population across the Wild Fields doubled from 14k in 6000BC to 24k in 5000BC.

Copper Age and Early Bronze Age Herders and Farmers

Between 4500BC and 3000BC, Balkan copper working and steppe horse herding transformed the region, with Budjak (Region I) and the Dniester-Bug coast (Region II) growing fastest [9][25][26]. Around Lakes Yalpug, Kagul, Katlabukh and Kitai in Budjak (Bolhrad, Ozernoye, Etulia VI, Plavni, Nagornoye, Suvorovo, Chervonyi Yar and Orlovka/Kartal), farmers of the Bolgrad-Aldeni branch of the Gumelnita Culture built villages of 20 to 40 wattle-and-daub houses (4200-3800BC, supporting 3,8k people in Region I by 4000BC) [9][27]. By 3500-3000BC, mingling between late Trypillia farmers and steppe kurgan herders of the Budjak Culture (known from Yamnaya and Usatovo burial mounds at Nerushai, Semenovka, Mayaki, Trapovka, Borisovka, Sarata and Tarutyne) lifted Budjak's population to 9,5k in 3500BC and 8,5k in 3000BC [1][9].

East of the Dniester Liman in Region II (3500-3000BC), stone-walled settlements of the Usatovo culture at

Usatovo and Mayaki near the Khadjibey and Kuialnyk estuaries presided over 14,5k inhabitants in 3500BC [14]. Together with Budjak (9,5k), the western coastal steppes (Regions I and II) held 24k people, or more than a third of the regional total. Further east in Regions III, IV and V (4000-3000BC), Sredny Stog horse-keepers at Dereivka (Region III) and the stone-walled Yamnaya fort at Mikhailovka (Region IV) anchored wagon-dwelling herders along the Ingulets, Molochnaya and Seversky Donets, raising the total population of the Wild Fields from 37k in 4000BC to 68k in 3500BC and 75k in 3000BC [24][26].

Middle and Late Bronze Age Villages

During the Middle Bronze Age Catacomb (2800-2000BC) and Babino (2200-1800BC) periods, herding populations grew steadily to 95k in 2000BC (8k in Budjak [I], 20k in NW Pontic [II], 24,5k in Right-Bank [III], 24k in Left-Bank [IV] and 18,5k in Middle Donets [V]) [1][9].

Prehistoric settlement reached its height in the Late Bronze Age under the Sabatinovka-Noua (1600-1200BC, Regions I-IV), Srubnaya (1600-1100BC, Regions IV-V) and Belozerka (1200-950BC) cultures [16][28][29]. In Budjak (Region I), more than 200 Sabatinovka-Coslogeni-Noua and Belozerka farming villages lined the Kogilnik, Sarata, Alkaliya and Danube lake shores (Zhovtneve/Karakurt, Ogorodnoye, Suvorovo, Tatarbunary, Diviziya, Tuzly, Shabo and Tudora), supporting 15k people in 1000BC [9][29]. In Region II (33k in 1000BC), stone-founded villages such as Sabatinovka, Voronovka II and Tashlyk combined cattle raising with millet farming [16]. Across the Right-Bank Ingul-Ingulets (Region III, 36k), Left-Bank Lower Dnieper (Region IV, 34k) and Donets-Azov basin (Region V, 27k), hundreds of stone-built villages (Zmiivka, Novokievka and Staryi Krym Ingul) flourished alongside the Srubnaya copper mines and bronze foundries of the Bakhmut basin (Kartamysh, Usovo Ozero and Kapitanovo in Region V), carrying the Wild Fields to 145k in 1000BC [3][28].

Classical Antiquity and the Chernyakhov Villages

Greek Ports in Budjak and on the Dniester-Bug Liman

From the mid-seventh century BC (~647BC), colonists from Miletus founded trading towns and farming villages across both Budjak (Region I) and the Dniester-Bug coast (Region II) [8][12][30]. In Budjak (27k in 500BC, 18k in 0AD and 29k in 300AD), Milesian settlers on the rocky western headland of the Dniester Liman (modern Bilhorod-Dnistrovskyi) founded Tyras (~600BC), which grew into a walled town of 6k inhabitants in 500-300BC surrounded by farming villages along the western shore of the estuary (Shabo, Zatoka, Mologa II, Pivdenne I and Belenkoye) [8][31]. At the south-western tip of Budjak overlooking the Danube crossing at the Kartal narrows near Reni, the fortified Geto-Scythian and later Roman river port of Orlovka/Kartal (ancient Aliobrix, with nearby Novoselskoye and Reni) guarded river traffic between the Danube Delta, the Prut and the Budjak steppe (1,5k - 2,2k inhabitants) [8][32].

In Region II (58k in 500BC, 40k in 0AD and 59k in 300AD), Berezan (~647BC, 2,5k) and Pontic Olbia (~600BC, a 50-55 ha walled town of 12k - 15k people) on the Bug-Dnieper Liman stood at the centre of more than 150 farming villages (*Shirokaya Balka*, *Kozyrka*, *Chertovatoe VII*, *Viktorovka I*, *Staraya Bogdanovka II* and *Didova Khata*) housing 20k to 25k Greco-Scythian grain growers [12][13][33]. On the eastern bank of the Dniester Liman opposite Tyras, Nikonion (Roksolany, 3k in 500-400BC) and its villages (Nadlimanskoye) linked Dniester trade to Koshary (Odessos, 1,4k at the Tiligul mouth) and the settlements around Odessa Bay (Zhevakhova Gora and Luzanovka) [15][34].

Royal Scythia, Late Scythian Forts and the Chernyakhov Peak

In the inland steppes of Regions III, IV and V, fifth- and fourth-century BC Royal Scythia centred on the Lower Dnieper Rapids and *Veliky Lug* (*Gerrhus*) [17][18]. At Kamenskoye Gorodishche (Region IV, enclosing 1,2k ha with its 32 ha Znamenka citadel and the 800-grave Mamai-Hora cemetery), a resident

population of 6k ironworkers and traders (plus 3,5k - 4,2k at Znamenka) lived opposite the great royal burial mounds of Region III (Chertomlyk, Tovsta Mohyla and Alexandropol) and Region IV (Solokha, Melitopol and Gaimanova Mohyla) [17][18]. In Region V (36k in 500BC), Scythian hillforts along the forest-steppe fringe (Velikaya Gomolsha and Koropovye Khutora) and camps on the Azov coast linked the Donets valley to the Dnieper heartland. Total population across the Wild Fields reached a Classical peak of 235k in 500BC (27k in Budjak [I], 58k in NW Pontic [II], 56k in Right-Bank [III], 58k in Left-Bank [IV] and 36k in Middle Donets [V]).

After the third-century BC Sarmatian advance pushed the total down to 165k at 0AD, settled farming recovered between 250BC and 200AD along both banks of the Lower Dnieper (Regions III and IV) around 16 stone forts (Zolota Balka, Krasny Mayak, Gavrilovka, Konsulovskoye, Nikolaevka-Kazatskoye, Lvove, Sabinka and Hannivka in Region III; Lyubimovka, Znamenskoye, Gornostaevka and Kairy in Region IV) [5][35][36][37]. At the same time, Roman garrisons at Tyras and Orlovka/Aliobrix (Region I) and Olbia (Region II) protected the coastal ports, lifting the total to 205k in 200AD [38].

Under the Chernyakhov Culture (250-375AD), hundreds of stone and timber farming villages spread across Budjak (29k in Region I at Bolhrad-Yalpug, Etulia, Chervonyi Yar, Kubei, Nagornoye, Belenkoye and Mologa II), the Dniester-Bug Liman (59k in Region II at Koblevo, Kamenka-Anchekrak and Ranconia), the Right-Bank Ingul-Ingulets and Rapids (62k in Region III at Bashmachka and Snihurivka), the Left-Bank Lower Dnieper (54k in Region IV) and the Donets valley (36k in Region V). Regional population reached an ancient peak of 240k in 300AD before the Hunnic invasion of 375AD cut the total to 150k in 400AD (18k in I, 36k in II, 38k in III, 35k in IV and 23k in V; Figure 2) [32][39][40].

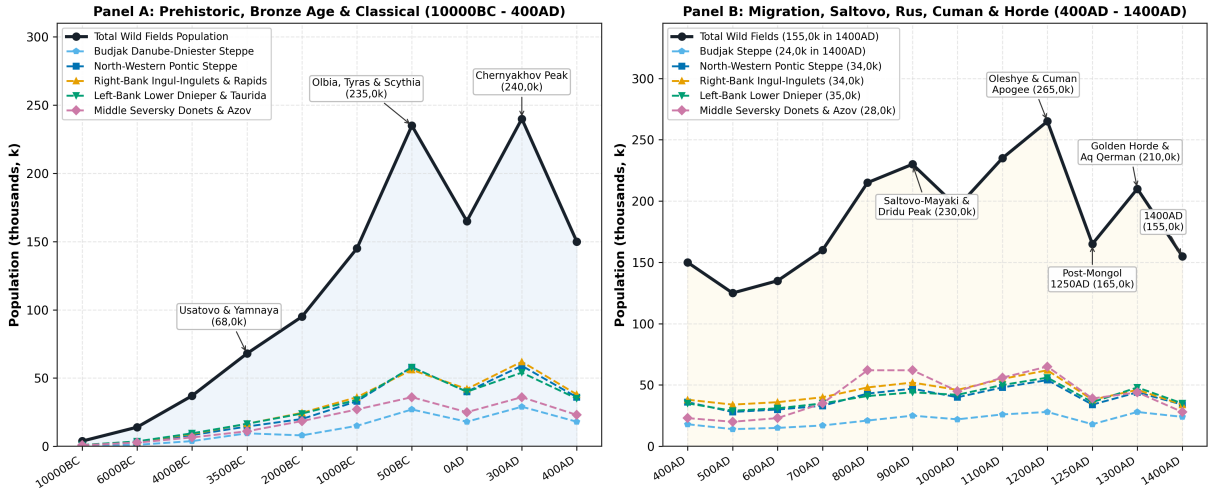


Figure 2. Total and regional population across the five divisions of the Wild Fields from 10000BC to 1400AD, showing the three ancient and medieval peaks under Classical Greek and Scythian trade (235k in 500BC), Chernyakhov farming (240k in 300AD) and Cuman-Rus commerce (265k in 1200AD).

Early and High Middle Ages

Steppe Nomads and Dnieper River Refuges

Between 400AD and 700AD, as Kutrigur and Utigur Bulgars, Avars and Khan Kubrat's Old Great Bulgaria crossed the open steppes, the population fell to 125k in 500AD (14k in Budjak [I], 28k in II, 34k in III, 29k in IV and 20k in V) before recovering to 135k in 600AD and 160k in 700AD [40][41]. Settled villages held on at the Lower Danube and Dniester crossings of Budjak (Orlovka/Kartal and Novoselskoye) and on the granite

islands of the Dnieper Rapids (Voloshskoye, Igren-Samar, Balka Kantsyrka and the c. 700AD princely camp at Voznesenka in Region IV) [41].

Bulgarian Villages in Budjak and the Saltovo-Mayaki Forts

From the eighth to the tenth century (750-965AD), the two opposite ends of the Wild Fields - Budjak (Region I) in the south-west and the Middle Seversky Donets (Region V) in the north-east - saw a strong revival of settled village life. After Asparukh crossed the *Onglos* in southern Budjak (680AD) and the First Bulgarian Empire established its northern frontier beyond the Danube, more than 70 eighth- to tenth-century Balkan-Danubian (*Dridu* or *Pliska-Preslav*) villages grew up around Lakes Yalpug, Kugurlui, Katlabukh, Kitai and Kagul (Suvorovo I-VI, Bogatoye, Ogorodnoye, Kholmskoye, Chervonyi Yar, Plavni, Etulia, Nagornoye, Safyany/Izmail and Orlovka/Kartal), guarded on the north by the earthen Lower Trajan's Wall (126 km from the Prut at Vadul-lui-Isac to Lake Sasyk) [9][10][42]. Settled by Bulgar, Slavic and Vlach farmers and fishermen, Budjak supported 21k people in 800AD and 25k in 900AD (~1,91 persons/km^2, the highest density in the Wild Fields).

Along the Lower Dniester, Southern Bug and Dnieper Rapids in Regions II, III and IV (47k, 52k and 44k in 900AD), East Slavic Tivertsi and Ulichi settlements (Belenkoye, Palanca, Peresechen Steppe, Chern, Voznesensk and Igren-Samar) traded down the rivers to the Black Sea [19][42]. At the same time, in Region V (62k in 800-900AD) under Khazar rule, Alanic and Proto-Bulgar settlers built more than 250 forts and open villages of the Saltovo-Mayaki culture along the Seversky Donets and Oskol valleys, led by the 120 ha Sidorovo complex (5k inhabitants), Mayaki-Tsaritsyno (3k), Sukhaya Gomolsha (2,6k), Mokhnach (2,5k), Netailovka (2,2k), Teplinskoye (2k), Zlivki (1,4k) and Krivaya Luka (1,3k) [6][23][43][44][45]. Across all five regions, the population of the Wild Fields rose to 215k in 800AD and 230k in 900AD.

Pechenegs, Kievan Rus Ports and the Cumans

Pecheneg raids and Prince Sviatoslav's 965AD campaign against Khazaria burned the Donets Saltovo forts and drove villagers from the open *Dridu* settlements in Budjak, cutting the regional total to 195k in 1000AD (22k in Budjak [I], 40k in II, 46k in III, 42k in IV and 45k in V) [6][46]. In the eleventh and twelfth centuries (1055-1223AD), however, trade between Kievan Rus, Byzantium and the Cuman (Polovtsian) tribes lifted the population to 235k in 1100AD and a pre-modern peak of 265k in 1200AD (28k in Budjak [I], 54k in NW Pontic [II], 62k in Right-Bank [III], 56k in Left-Bank [IV] and 65k in Middle Donets [V]) [19][46][47].

At the Danube and Dniester mouths in Region I (28k in 1200AD), Byzantine, Galician-Rus (*Berladniki*), Vlach and Cuman trade revived Maurocastro/Belgorod (ancient Tyras, 3k - 3,5k), Kiliia (2,5k), Smil/Izmail (1,8k), Reni (1k) and Orlovka/Kartal (1,2k) [10][11]. At the head of the Dnieper estuary in Region IV, the Kievan Rus port of Oleshye (3,5k in 1200AD) served as Kiev's sea outlet to Constantinople, paired upstream with the Khortytsia ferry and *Brodnik* settlement of Protolche (2,2k) and the 25 ha town of Igren-Samar at the Samara mouth (2,8k) [19][48]. Along the Middle Seversky Donets in Region V, Cuman-Alanic winter towns grew up at Sharukhan (4,5k), Sugrov (3,2k) and Balin (2,8k), while more than 2,5k Polovtsian stone statues (*kamennye baby*) and rich burial mounds (such as the Chingul Kurgan on the Molochnaya in Region IV) marked twelfth- and early thirteenth-century clan pastures (Figure 3) [22][46].

The Golden Horde and Late Medieval Ports

Mongol Conquest and Town Revival

The Mongol campaigns of 1223AD (Battle of the Kalka River in Region V) and 1238-1240AD cut the population of the Wild Fields from 245k in 1240AD to 165k in 1250AD (18k in Budjak [I], 34k in II, 38k in III, 36k in IV and 39k in V) [47][48]. Under the western Golden Horde (Prince Nogai, d. 1299AD; Khan

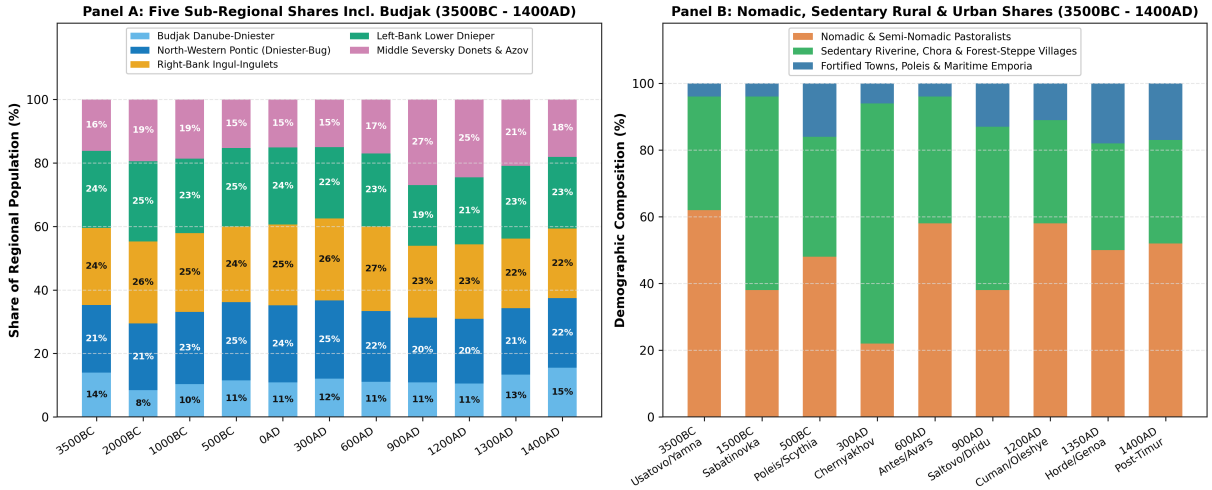


Figure 3. Regional population shares across the five divisions of the Wild Fields (left) and the changing balance between coastal and river towns, farming villages and mobile steppe herders from 3500BC to 1400AD (right).

Toqta, 1291-1312AD; and Khan Uzbek, 1313-1341AD), however, trade and town life revived, carrying the total back to 210k by 1300AD [20][21][49].

Under Nogai and Toqta, Budjak (Region I, 28k in 1300AD) became the main western port district of the Golden Horde. At the Dniester mouth, Aq Qerman (Maurocastro or Cetatea Alba) minted its own coins, housed Genoese, Armenian, Greek, Jewish, Moldavian and Tatar merchant quarters and reached 15k - 16k inhabitants at its height in 1300-1345AD [10][11]. On the Chilia arm of the Danube Delta, the port of Kiliia/Lycostomo (where Antonio di Ponzò's 1360-1361AD notary registers record shipments of grain, wax, salt and wine to Pera and Genoa) held 6,5k - 8k people, flanked by Smil/Izmail (3,5k - 4,5k), Reni (1,3k) and the inland Golden Horde town of Kara-Bunar/Tatarbunary (1,1k) [11]. In the *Veliky Lug* floodplain of the Left-Bank Lower Dnieper (Region IV, 48k in 1300AD), Uzbek and Jani Beg built the 300 ha open town of Velyki Kuchugury (with a mosque, minaret, bathhouse, water pipes and 5,5k - 6k inhabitants), alongside Mamai-Surka, Konski Vody/Mechetna Mohyla (4k) and Oleshye (2k), while Italian sea charts marked Porto Pisano (Belosaraiskaya, 1,5k) on the Azov coast of Region V [20][21].

The Black Death, Timur's Campaign and the 1400AD Benchmark

Between 1346AD and 1400AD, the inland steppes of Regions II, III, IV and V lost much of their population to the Black Death (1346-1348AD), the civil wars of the Golden Horde (1359-1380AD), Grand Duke Algirdas's campaign at the Battle of Blue Waters (1362AD, on the Syniukha River in Region III), Timur's 1395AD campaign (which destroyed Kuchugury, Mamai-Surka, Konski Vody and the Donets towns) and the Battle of the Vorskla River (1399AD) [21][49]. Budjak (Region I) and the ports of Region II held up much better as the Principality of Moldavia (under Petru I Musat and Alexander the Good in the 1390s - 1400AD) fortified Cetatea Alba/Aq Qerman (12k in 1400AD), Kiliia/Chilia (6k) and Smil/Izmail (3,5k), while Grand Duke Vytautas of Lithuania built fortified customs ports at Dashov/Kara-Kerman (Ochakiv, 2,8k in 1400AD), Khadzhibey/Koczubij (Odessa Bay, 2,5k), Sokolets, Balyklei and Tyagin [11].

By 1400AD the population of the Wild Fields stood at 155k: 24k in Budjak (15,5% of the total on 5,5% of the land area), 34k in Region II, 34k in Region III, 35k in Region IV and 28k in Region V (Figure 4).

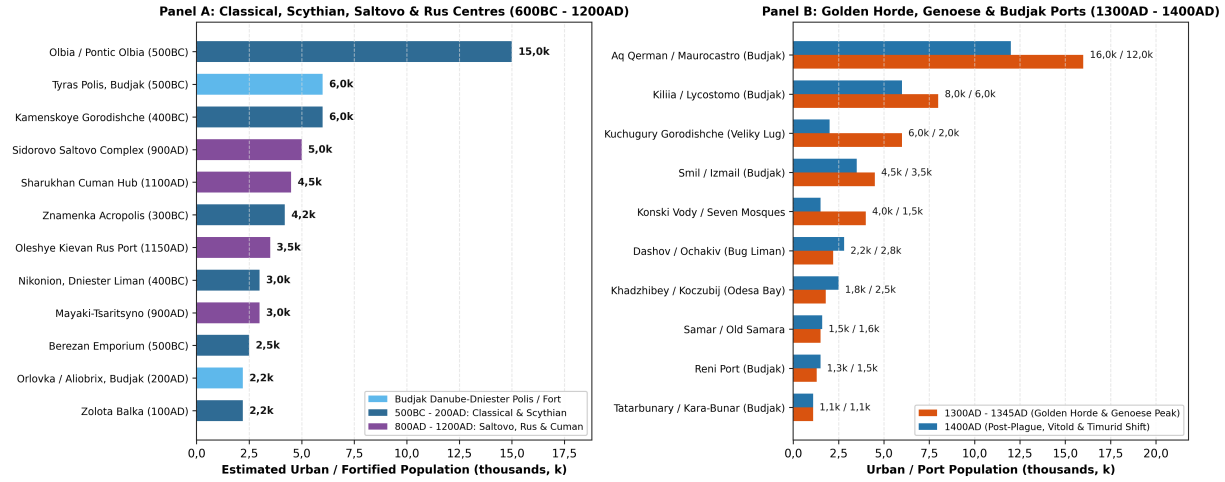


Figure 4. Leading towns and river forts of the Wild Fields in Classical antiquity (500BC - 200AD, left) and the Middle Ages (800 - 1400AD, right), comparing Pontic Olbia (15k), Tyras (6k) and Kamenskoye (6k) with Aq Qerman (16k), Kiliia (8k), Kuchugury (6k) and Sidorovo (5k).

Regional Population Summary

Table 2 sets out the complete benchmark population series for the Wild Fields and Budjak from 10000BC to 1400AD.

Date	I: Budjak	II: NW Pontic	III: R-Bank	IV: L-Bank	V: Donets	Total	Historical & Archaeological Context
10000BC	0,3k	0,8k	0,95k	0,95k	0,8k	3,8k	Epigravettian bison & horse hunters (Anetovka II, Amvrosievka)
8000BC	0,55k	1,45k	1,7k	1,7k	1,4k	6,8k	Kukrek & Grebeniki Mesolithic camps & rapids cemeteries (Mirnoe [Budjak], Vasylivka)
6000BC	1,1k	3,1k	3,5k	3,6k	2,7k	14k	Bug-Dniester, Sursko-Dnieper & Danube-Lakes Early Neolithic villages
5000BC	1,9k	5,3k	6k	6,2k	4,6k	24k	Dnieper-Donets II & Mariupol Neolithic collective cemeteries (Vovnigi, Mariupol)
4000BC	3,8k	8,2k	9k	9,5k	6,5k	37k	Bolgrad-Aldeni (Gumelnita) in Budjak; Sredny Stog & Dereivka horse-keepers

3500BC	9,5k	14,5k	16,5k	16,5k	11k	68k	Budjak Kurgan & Usatovo coastal settlements; Early Mikhailovka Copper Age peak
3000BC	8,5k	14,5k	19k	19k	14k	75k	Yamnaya (Pit-Grave) wagon pastoralism (Nerushai, Mikhailovka stone fort)
2000BC	8k	20k	24,5k	24k	18,5k	95k	Catacomb (Ingul/ Donets) & Babino Middle Bronze Age pastoralism
1000BC	15k	33k	36k	34k	27k	145k	Sabatinovka-Coslogeni (Budjak), Srubnaya & Belozerka Late Bronze village expansion
500BC	27k	58k	56k	58k	36k	235k	Classical Greek Poleis (Tyras [Budjak], Olbia, Nikonion) & Royal Scythia (Kamenskoye)
0AD	18k	40k	42k	40k	25k	165k	Sarmatian migrations & early Late Scythian Lower Dnieper stone forts
200AD	24k	50k	52k	48k	31k	205k	Roman-allied Tyras, Aliobrix [Budjak] & Olbia; 16 Late Scythian forts
300AD	29k	59k	62k	54k	36k	240k	Chernyakhov farming & herding peak across Budjak, Liman & Dnieper valleys
400AD	18k	36k	38k	35k	23k	150k	Post-375AD Hunnic destruction of Chernyakhov villages & coastal towns
500AD	14k	28k	34k	29k	20k	125k	Early Medieval Kutrigur/Utigur nomadism; Antae/ Penkovka at Dnieper Rapids
600AD	15k	30k	36k	31k	23k	135k	Avar & Old Great Bulgaria steppe transit; Danubian & Rapids settlements

700AD	17k	33k	40k	35k	35k	160k	Asparukh's Onglos in Budjak; Voznesenka princely camp; early Khazar rule
800AD	21k	43k	48k	41k	62k	215k	Dridu (First Bulgarian Empire) villages in Budjak; Saltovo-Mayaki growth on Donets
900AD	25k	47k	52k	44k	62k	230k	Saltovo-Mayaki (Sidorovo 5k) & Budjak Dridu peak; Ulichi & Tivertsi towns
1000AD	22k	40k	46k	42k	45k	195k	Post-965AD Saltovo collapse; Pecheneg steppe dominance; Kievan Rus Oleshye
1100AD	26k	48k	55k	50k	56k	235k	Polovtsian winter towns (Sharukhan) & Rus Oleshye, Protolche & Berladniki
1200AD	28k	54k	62k	56k	65k	265k	High Medieval Cuman-Rus trade & settlement peak (Maurocastro, Oleshye)
1240AD	26k	50k	57k	52k	60k	245k	Pre-Mongol eve (after 1223AD Battle of Kalka River)
1250AD	18k	34k	38k	36k	39k	165k	Post-1240AD Mongol conquest drop across inland steppes
1300AD	28k	44k	46k	48k	44k	210k	Golden Horde & Genoese town revival (Aq Qerman 16k & Kiliia 8k in Budjak; Kuchugury 6k)
1400AD	24k	34k	34k	35k	28k	155k	Post-Plague & 1395AD Timurid steppe drop; Moldavian-Genoese Budjak & Lithuanian ports

Table 2. Complete population series across the five regions of the Wild Fields and Budjak at 26 benchmark dates, paired with the main archaeological cultures, nomadic migrations and trading states from the Epigravettian to 1400AD.

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